

General Knowledge

Indian Economy

STUDY NOTES

Money Supply

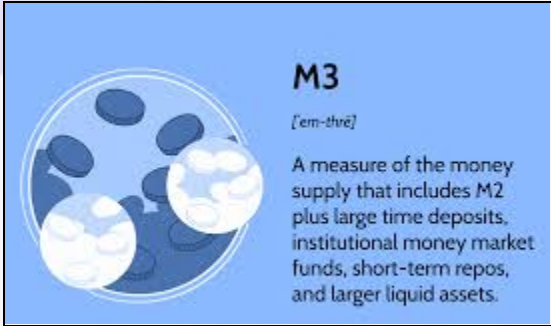


Money Supply

The **money supply** refers to the total amount of money circulating in an economy at a specific time. It **includes all physical currency (coins and notes)** and various types of bank deposits that people can use for transactions or investments.

Components of Money Supply

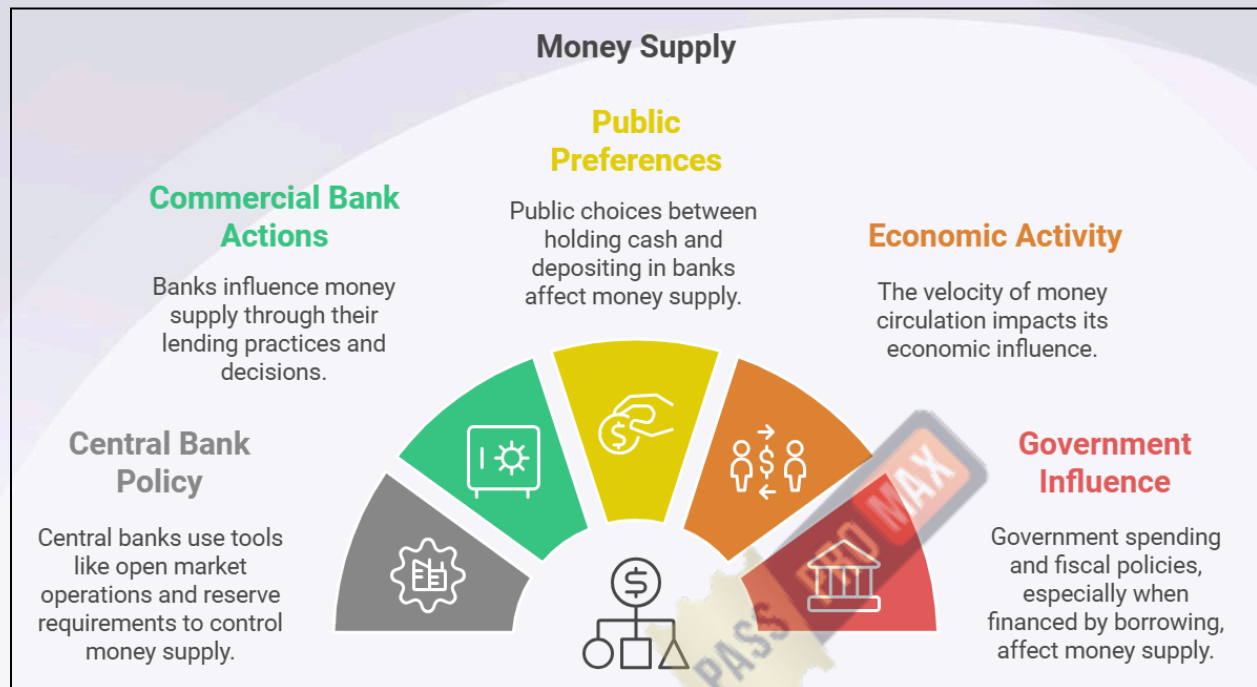
Components	Description
M1 (Narrow Money)  M1 [em-'wan] The total amount of physical currency and coin, demand deposits, traveler's checks, and other checkable deposits circulating in an economy.	
Currency in Circulation	Physical cash (coins and paper money) held by the public.
Demand Deposits	Funds held in checking accounts can be withdrawn immediately.
Traveller's Checks	Financial instruments used for payments.
M2 (Broad Money)  M2 [em-'yu] A measure of the money supply that includes cash, checking deposits, and non-cash assets that can easily be converted into cash.	
M1 Components	All components of M1.
Savings Deposits	Funds held in savings accounts, often with withdrawal restrictions.

Time Deposits	Fixed-term deposits like certificates of deposit (CDs).
Money Market Mutual Funds	Funds that invest in short-term debt securities.
M3 (Broadest Money)	
	
M2 Components	All components of M2.
Large Time Deposits	Time deposits with large denominations.
Institutional Money Market Funds	Money market funds held by institutions.
M4 (Broad Money)	
Includes M3, plus additional elements like commercial paper and T-Bills	

Measures of Money Supply

- The **money supply is measured using various aggregates, denoted as M1, M2, M3, and so on.** The choice of measure depends on the liquidity and accessibility of the assets included.
- The commonly used measures are M1 and M2:
- **M1:**
 - $M1 = C + D$ (Currency in Circulation + Demand Deposits)
- **M2:**
 - $M2 = M1 + S + T + M$ (M1 + Savings Deposits + Time Deposits + Money Market Mutual Funds)
- **Other measures, such as M3,** include broader categories of financial assets and are used in specific economic contexts.

Factors Influencing Money Supply



Sources of Money Supply

The sources of money supply involve both the central bank (typically the Reserve Bank of India, RBI, in India) and commercial banks. The key sources include:

- **Central Bank (RBI) Actions:**

- The RBI has the authority to issue currency notes and coins, contributing to the physical component of the money supply (M0).
- Through monetary policy tools, the RBI influences the money supply by adjusting interest rates, conducting open market operations, and setting reserve requirements.

- **Commercial Banks:**

- Commercial banks create money through the process of lending. When banks make loans, they effectively create new deposits in the borrower's account, increasing the overall money supply.
- Banks also hold reserves with the central bank, affecting the high-powered money component (M0).

RBI Measures of Money Supply

The Reserve Bank of India (RBI) classifies the money supply into four monetary aggregates (M1, M2, M3, and M4) based on the components included in each measure. These classifications help the RBI analyze and manage the money supply effectively.

Expansion of Money Supply

The expansion of the money supply can occur through various mechanisms:

- **Bank Lending:** When banks lend money, they create deposits in the borrower's account, contributing to the expansion of the money supply.
- **Central Bank Operations:** Open market operations by the central bank, involving the buying or selling of government securities, can impact the money supply.
- **Government Spending:** Government expenditure, especially when financed through borrowing, can contribute to an expansion of the money supply.
- **Decrease in Reserve Requirements:** If the central bank lowers reserve requirements for commercial banks, it increases their capacity to lend, contributing to an expansion of the money supply.
- **Foreign Exchange Operations:** Foreign exchange transactions by the central bank can also impact the domestic money supply.

Money Multiplier

The **Money Multiplier** is the ratio that measures how much the money supply in an economy increases for every unit of reserve money (or base money) created by the central bank. It indicates the potential of the banking system to expand the total money supply.

$$\text{Money Multiplier} = \frac{\text{Total Money Supply}}{\text{Reserve Money (Base Money)}}$$

Alternatively, it can also be expressed as:

$$\text{Money Multiplier} = \frac{1}{\text{Reserve Ratio (RR)}}$$

Where:

- **Reserve Ratio (RR):** The fraction of deposits that banks are required to hold as reserves.

Key Features

- **Amplification Effect:** A small amount of reserve money leads to a larger total money supply due to the lending activities of banks.
- **Role of Banks:** Banks lend a portion of deposits, and the recipients redeposit that money, enabling further loans.
- **Inverse Relationship with Reserve Ratio:** Higher reserve requirements lead to a lower money multiplier, reducing the money supply.

Factors Influencing the Money Multiplier

1. **Reserve Ratio (RR):** A higher reserve ratio decreases the money multiplier and vice versa.
2. **Currency to Deposit Ratio (C/D):** If people prefer holding cash instead of deposits, the money multiplier decreases.
3. **Bank Lending Practices:** More aggressive lending increases the money multiplier, while conservative lending reduces it.
4. **Public Behavior:** Greater trust in banks leads to higher deposits, enhancing the multiplier effect.
5. **Central Bank Policies:** Policies on reserve requirements and interest rates significantly influence the money multiplier.

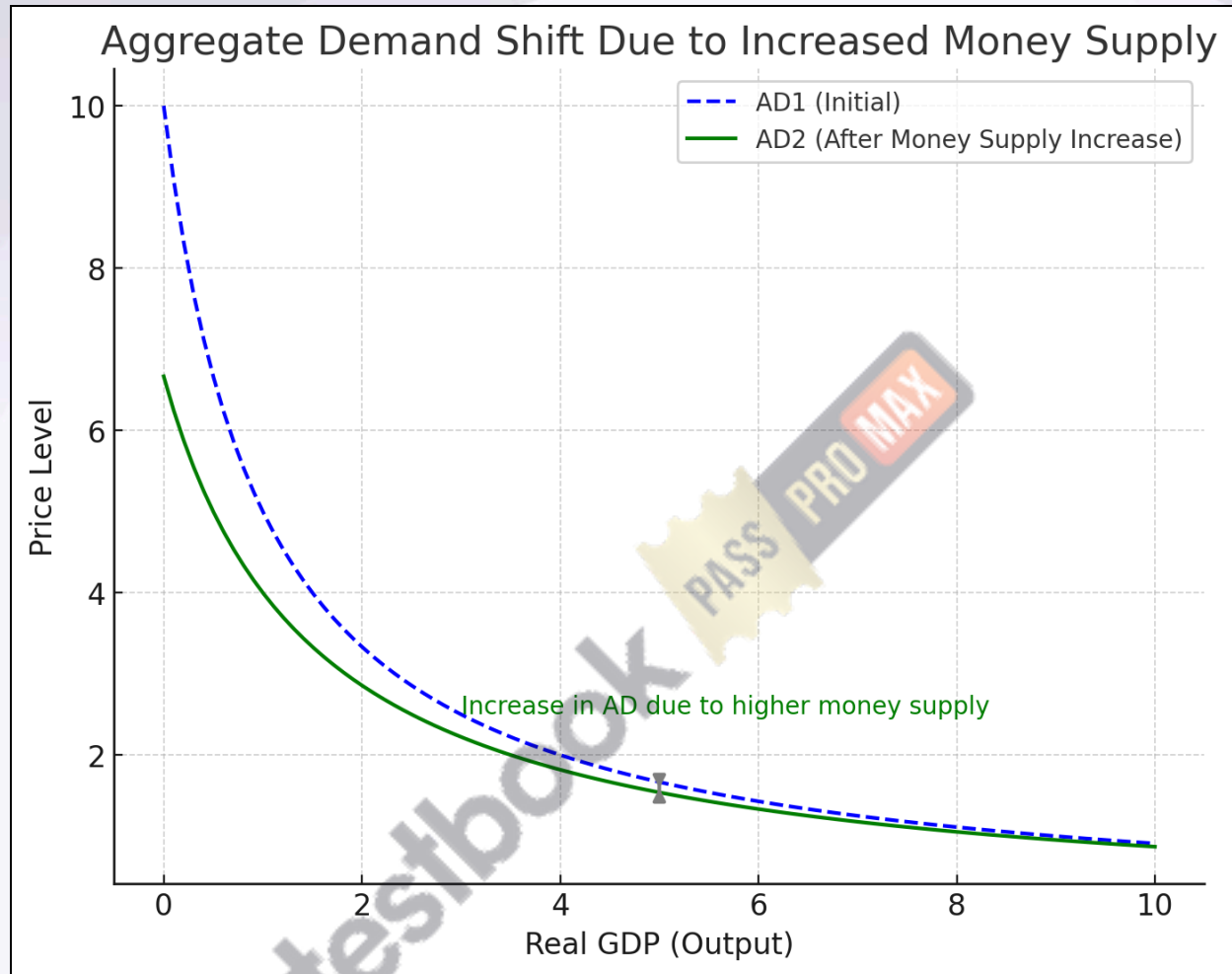
Measures and Theories Related to Money Supply

Keynesian Perspective

- Keynesians view money supply as one of several tools for managing economic output and unemployment.
- They argue that increasing the money supply can stimulate demand by lowering interest rates, and encouraging borrowing, and spending.
- In a recession, Keynesians advocate for increasing the money supply to spur demand. However, they caution that this approach might not be as effective in certain conditions,

such as a liquidity trap (where low interest rates do not stimulate borrowing).

- Keynesian economics emphasizes fiscal policy more than monetary policy but acknowledges the role of money supply in managing short-term economic fluctuations.

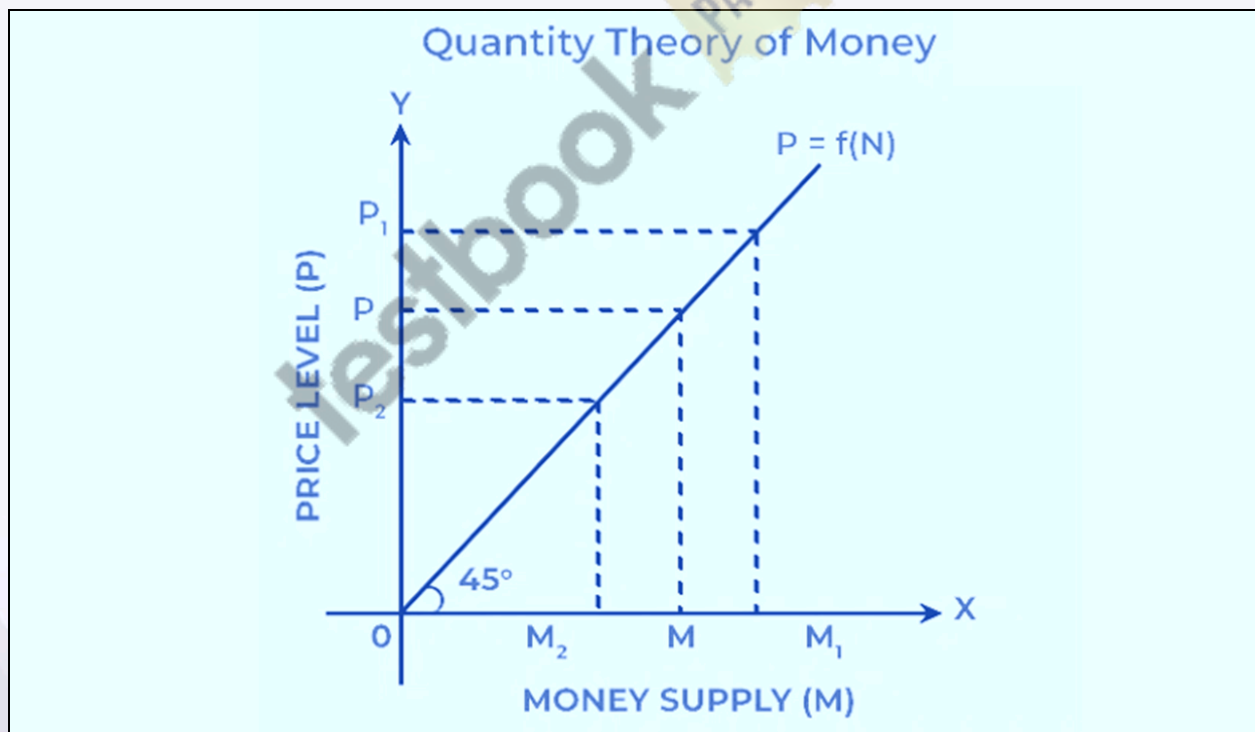


Here is a simple graph illustrating the effect of an increased money supply on aggregate demand (AD).

- AD1 (Initial) shows the original aggregate demand curve before the increase in money supply.
- AD2 represents the new aggregate demand curve after the increase in money supply, shifted to the right.
- The arrow indicates this rightward shift, showing that an increase in money supply boosts aggregate demand, leading to a higher real GDP (output) and price level in the short term.

Monetarist Perspective

- Monetarists, particularly Milton Friedman, emphasize the role of money supply as the primary driver of economic performance and inflation.
- According to the Quantity Theory of Money, the money supply is directly proportional to the price level in the long term. This theory is summarized by the equation $MV = PQ$ or $MV = PQ$:
 - MMM: Money Supply
 - VVV: Velocity of Money (how frequently money is used in transactions)
 - PPP: Price Level
 - QQQ: Real Output (quantity of goods and services produced)
- Monetarists believe that increasing the money supply too quickly leads to inflation, while a stable and controlled money supply is essential for economic stability.
- They generally argue for fixed, rule-based increases in the money supply rather than discretionary interventions by central banks.



Here's an illustration of the Quantity Theory of Money equation ($MV = PQ$) with labels explaining each term:

- M (Money Supply): The total currency and liquid assets in circulation.
- V (Velocity of Money): How frequently each unit of currency is used in transactions over a period.
- P (Price Level): The average level of prices for goods and services.
- Q (Real Output): The total quantity of goods and services produced.

The arrow demonstrates that if M (money supply) increases while V (velocity) and Q (output) remain constant, then P (price level) will increase. This shows the direct relationship between money supply and price level in this model.

Impact on Inflation and Economy

- **Inflation** is the rate at which the general price level of goods and services rises, leading to a decrease in the purchasing power of money.
- When the money supply increases faster than the economy's production of goods and services (real output), inflation tends to occur. With more money available, people can afford to spend more, which drives up demand and, consequently, prices.
- This relationship is often discussed through the Quantity Theory of Money, which suggests that if the velocity of money (V) and real output (Q) are stable, an increase in money supply (M) directly raises the price level (P).

Real-World Example: Zimbabwe in the late 2000s experienced hyperinflation when its central bank printed excessive amounts of money to fund government spending, leading to severe inflation. This caused prices to soar, and the currency eventually lost nearly all value.